

Project Management for Bioinformatics

Bioinformatics Workshop · Tanzania 2026

Adapted from the Data Carpentry lesson (CC-BY 4.0)

Today's roadmap

1

Data tidiness

What metadata to collect,
and how to structure it so
computers can use it

2

Project planning

What goes to the sequencing
facility, what comes back,
and how to store it

3

Public data on the SRA

Finding, inspecting and
downloading published
sequencing data

Part 1 · Data tidiness

What metadata should I collect, and how should I structure it?

Metadata: the data about your data

Without information about your samples, your samples are useless.

Experiment data

Spreadsheets of whatever you measured in your study

Lab notebook

How you conducted the experiments

Submission sheet

Sample names, DNA concentration, etc. for the facility

Prep notes

Library prep and sequencing type, e.g. paired-end Illumina

DISCUSS WITH A NEIGHBOR

What data and information did you generate before you sent your DNA/RNA off for sequencing?

Was it more than you expected?

Notes, unique IDs and standards

1

Date everything

Put dates on notebook pages, on the samples themselves, and in every record about them.

2

Use unique identifiers

Give each sample or dataset a name that exists only for it. Dates help: the same sample is rarely processed twice on one day.

3

Follow a metadata standard

Check for standards, like the Digital Curation Centre list and the Genomic Standards Consortium. If none exist, record *the minimum someone would need to use your data without talking to you*.

The cardinal rules of spreadsheets

1 Leave raw data raw

Never edit the original

2 Samples in rows

Each observation gets its own row

3 Variables in columns

e.g. strain, DNA_concentration

4 No spaces in headers

library_prep or LibraryPrep, not
'library prep method'

5 One value per cell

Species and strain in separate columns

6 Export as CSV

Text-based, readable anywhere,
required by repositories

One value per cell: an example

Hard to sort or filter

sample	organism
S1	E. coli K12
S2	E. coli REL606

Tidy

sample	species	strain
S1	E. coli	K12
S2	E. coli	REL606

Part 2 · Project planning

What does the facility need, what comes back, and where does it live?

Before you start: questions to ask

?

What challenges will you face working with a large sequence dataset?

?

How will you save and share your sequence files?

?

How can you be sure your raw data haven't been corrupted?

?

Where will you analyze the data: what software, what computers?

Sending samples to the facility

Common errors in submission sheets

- Inconsistent sample-ID formats; spaces or slashes in names
- Capitalization changes within a column
- Special characters (μ) in column headers
- Decimal precision varies down a column
- Several date formats in one sheet
- Hidden ones: duplicate names or barcodes, missing values

EXERCISE

Open `sample_submission.txt` (tab-delimited).

1. What errors can you spot?
2. How could the naming be improved?
3. Which errors are hard to see, and how could you test for them?

What comes back from a genomics facility

`_R1 / _R2`

Read 1 and read 2: forward and reverse reads of the same fragment (paired-end)

`.gz`

gzip-compressed files, to save disk space

`sample_id`

The key that links file names back to your submission sheet (e.g. VLOOKUP)

1.1 TB

total size of the example results files

Check file sizes after every transfer

Storing your raw data

1

Accessible

You and at least your lab head
can reach it

2

Redundant

Backed up in two physically
separate locations

3

Raw stays raw

Never modify the stored
original

Where to put it

Campus HPC or data-storage facility (ideal) · two external hard drives kept in different places · cloud storage (Amazon S3, Azure, Google Cloud) · Open Science Framework (free, files up to 5 GB)

Key points

1

Metadata lets you, and others, work with your data

2

Structure tabular data so computers can use it

3

Data sent to the facility needs the same care

4

Keep raw sequence data raw, and backed up twice